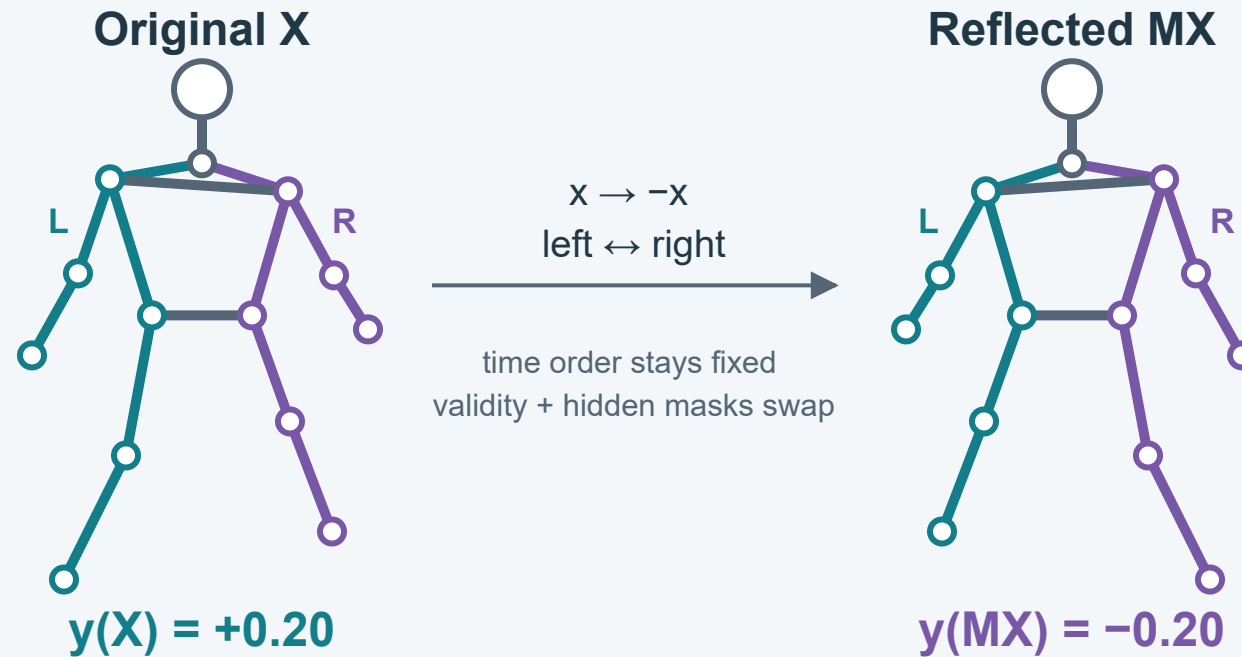


A known geometric transformation gives a testable prediction

The body may walk asymmetrically; reflecting its measurement should still exchange left and right.

A ANATOMICAL REFLECTION

Coordinate sign, joint identities, and masks transform together



Illustrative body and values; the implemented schema contains 33 landmarks.

B WHAT THE TARGET MEASURES

Paired motion from observed coordinates

For each left/right landmark pair:
uL and uR are median movement speeds
over the same valid time transitions.

$$\text{pair contrast} = (uL - uR) / (uL + uR + \epsilon)$$

Average the five accepted pair contrasts.
Use shoulder, knee, ankle, heel, and toe pairs.
Each pair needs at least eight valid transitions.
All five pairs must be usable.

Positive y: positive mean left-minus-right contrast.
Opposing differences between pairs can cancel.

Token equivariance

$Z(MX) \approx P Z(X)$ exchanges joint positions.
Feature channels are assumed to stay fixed.
The explicit loss has synthetic evidence only.

Exact oddness at the readout

$g(X) = [h(X) - h(MX)] / 2$
Then $g(MX) = -g(X)$, even before training.
Predictive benefit requires a separate test.

Constant-feature control

Constant nonzero tokens can match perfectly.
They cannot distinguish clips, so we also
measure feature variation and held-out R^2 .